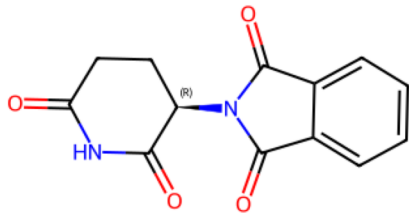
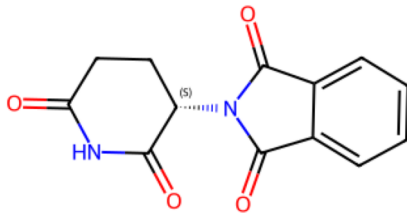


(R)-thalidomide $\chi = +1.04$



(S)-thalidomide (mirror) $\chi = -1.04$



chirality completeness on a real molecule
(19 heavy atoms)

0(3)-invariant readouts, M vs mirror M':

all 171 pairwise distances

$\max|\Delta| = 0e+00$

all pairwise dot products (Gram)

$\max|\Delta| = 0e+00$

$\Rightarrow$ an even readout is blind

parity-odd 0o pseudoscalar χ :

$\chi(M) = +1.037$, $\chi(M') = -1.037$

sum = +0e+00 (exact sign flip)

RDKit CIP: R $\rightarrow$ S